

ECOSPHERE – GREEN COMPUTING

A Mini Project Report

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Science in Computer Science (B.Sc. CS)

Submitted by

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CERTIFICATE

This is to certify that the project entitled “Ecosphere – Green Computing” is a Bonafide work carried out by KANSUL MAHARIBA S in partial fulfillment of the requirements for the award of the Bachelor of Science in Computer Science (B.Sc. CS) during the academic year 2025–2026.

The project has been completed under the guidance and supervision of the Department of Computer Science and has not been submitted to any other institution for the award of any degree or diploma.

Guide Signature: _____

Head of the Department: _____

Principal: _____

ACKNOWLEDGEMENT

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I extend my heartfelt thanks to our Principal, Head of the Department, and my Project Guide for their valuable guidance, encouragement, and continuous support throughout the development of this project.

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Finally, I thank everyone who directly or indirectly contributed to the successful completion of my project, “Ecosphere – Green Computing.”

TABLE OF CONTENTS

S. No.| Contents| Page No.

1| Certificate| 2

2| Acknowledgement| 3

3| Table of Contents| 4

4| Abstract| 5

5| Introduction| 6

6| Objectives| 7

7| Technologies Used| 8

8| Project Modules| 9

9| Screenshots| 10

10| Conclusion| 11

11| References| 12

ABSTRACT

Ecosphere – Green Computing is a web-based project developed to create awareness about environmental sustainability and energy-efficient computing practices. The website provides information on climate change, green technology, and green computing, encouraging users to adopt eco-friendly habits in their daily lives.

The project is developed using HTML5, CSS3, and JavaScript to create a responsive and interactive user interface. It includes a Hero Section, Climate Change, Green Technology, Green Computing, a Carbon Footprint Calculator, and a Contact Section. The Carbon Footprint Calculator allows users to estimate their carbon emissions based on energy consumption, helping them understand the importance of reducing their environmental impact.

The main objective of this project is to promote sustainable technology, reduce electronic waste, encourage energy conservation, and spread awareness about protecting the environment. The website is simple, user-friendly, responsive, and suitable for educational purposes.

INTRODUCTION

Green computing is the practice of designing, using, and managing computer systems in an environmentally responsible manner. It focuses on reducing energy consumption, minimizing electronic waste, and promoting sustainable use of technology.

Ecosphere – Green Computing is a web-based project developed to create awareness about climate change, environmental protection, and eco-friendly technologies. The website provides useful information on green technology, green computing practices, and sustainable solutions that help protect the environment.

The project is developed using HTML5, CSS3, and JavaScript, providing a modern, responsive, and user-friendly interface. It also includes a Carbon Footprint Calculator, allowing users to estimate their carbon emissions and understand the importance of reducing energy consumption.

The primary goal of this project is to educate users about environmental sustainability and encourage them to adopt green computing practices for a cleaner, healthier, and more sustainable future.

OBJECTIVES

The main objectives of the Ecosphere – Green Computing project are:

- To create awareness about climate change and environmental sustainability.
- To promote the use of green technology and renewable energy sources.
- To encourage green computing practices to reduce energy consumption.
- To reduce electronic waste (E-Waste) through responsible usage and recycling.
- To educate users about the importance of conserving natural resources.
- To provide a simple and interactive Carbon Footprint Calculator for estimating carbon emissions.
- To develop a responsive and user-friendly website using HTML5, CSS3, and JavaScript.
- To inspire individuals to adopt eco-friendly habits and contribute towards a sustainable future.

TECHNOLOGIES USED

Front-End Technologies

HTML5

HTML5 is used to create the structure and content of the website. It provides semantic elements that improve readability and organization.

CSS3

CSS3 is used to design and style the website. It provides colors, layouts, animations, responsive design, and an attractive user interface.

JavaScript

JavaScript is used to add interactivity and dynamic functionality. It is used for the Welcome Popup, Carbon Footprint Calculator, and other interactive features.

Development Tool

Visual Studio Code (VS Code)

Visual Studio Code is used as the source code editor for developing and managing the project.

Web Browser

Google Chrome

Google Chrome is used to run, test, and debug the website during development.

PROJECT MODULES

1. Hero Section

The Hero Section is the homepage of the website. It introduces the Ecosphere – Green Computing project with an attractive banner, welcome message, and a "Start Journey" button that guides users to explore the website.

2. Climate Change

This module explains the causes and effects of climate change. It creates awareness about global warming, greenhouse gas emissions, and the importance of protecting the environment.

3. Green Technology

This section presents eco-friendly technologies such as Solar Energy, Wind Energy, Electric Vehicles, and Smart Agriculture. It highlights how these technologies help reduce pollution and support sustainable development.

4. Green Computing

The Green Computing module explains energy-efficient computing practices, responsible use of electronic devices, and methods to reduce electronic waste (E-Waste).

5. Carbon Footprint Calculator

This interactive module allows users to estimate their carbon emissions by entering energy consumption values. It helps users understand their environmental impact and encourages energy conservation.

6. Content Section

The Contact section provides communication details and encourages users to connect for feedback, suggestions, or queries related to the project.

SCREENSHOTS

Figure 1: Hero Section

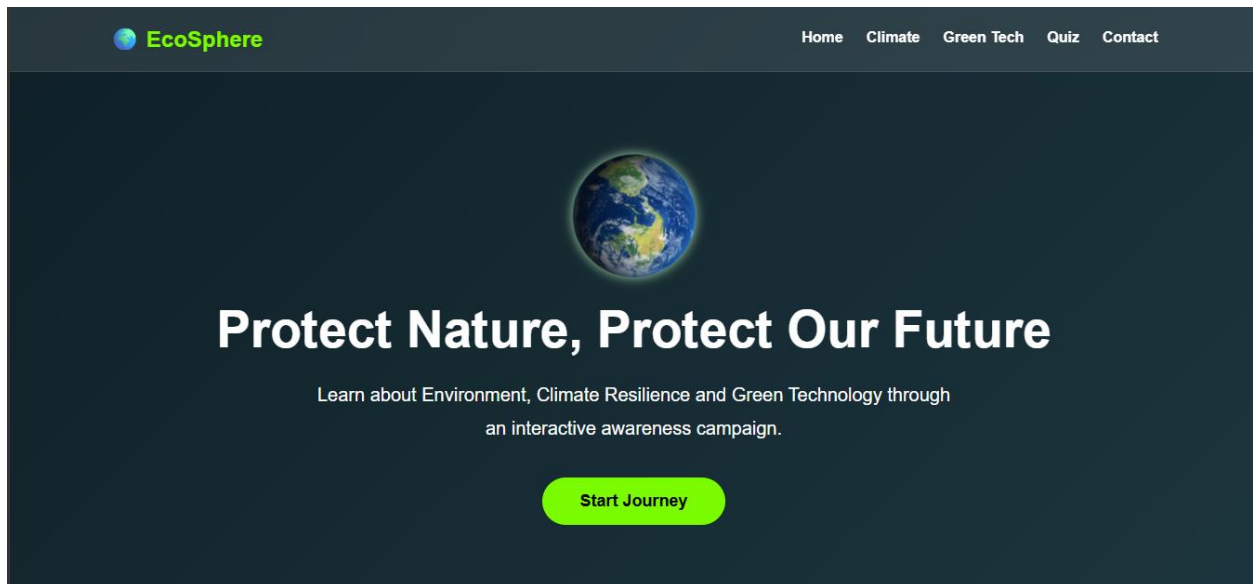


Figure 2: Climate Change

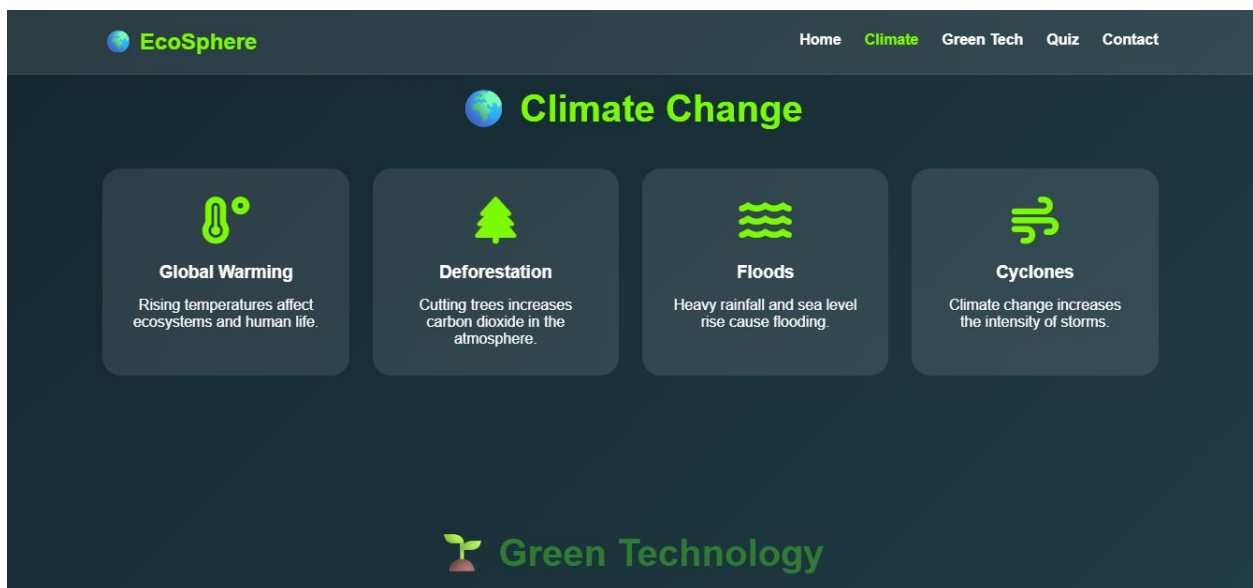


Figure 3: Green Technology

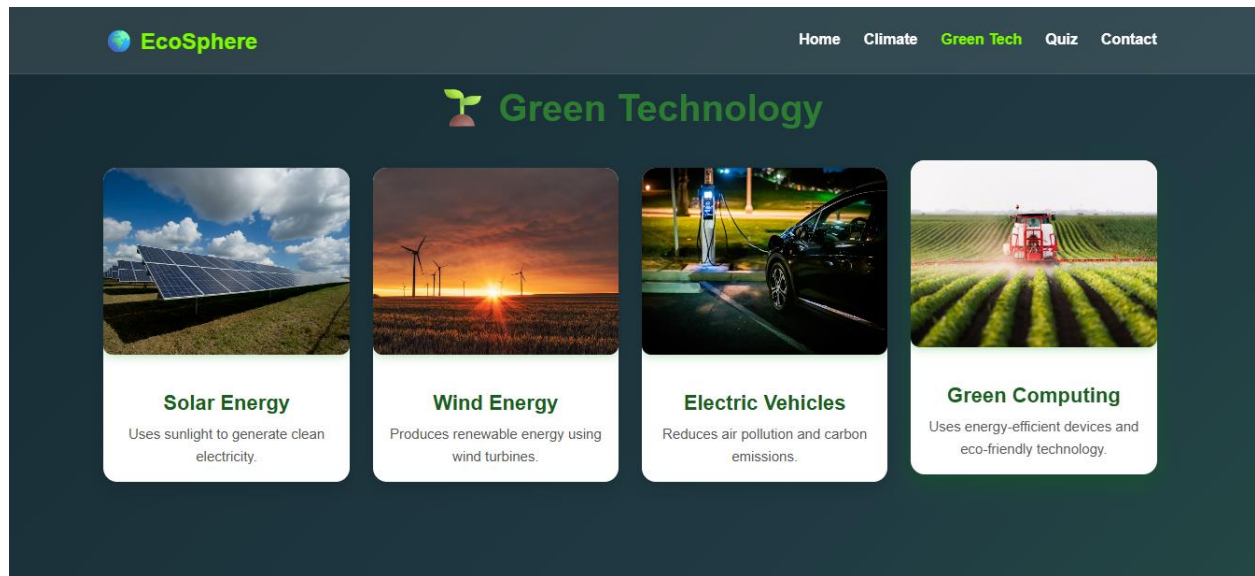


Figure 4: Green Computing

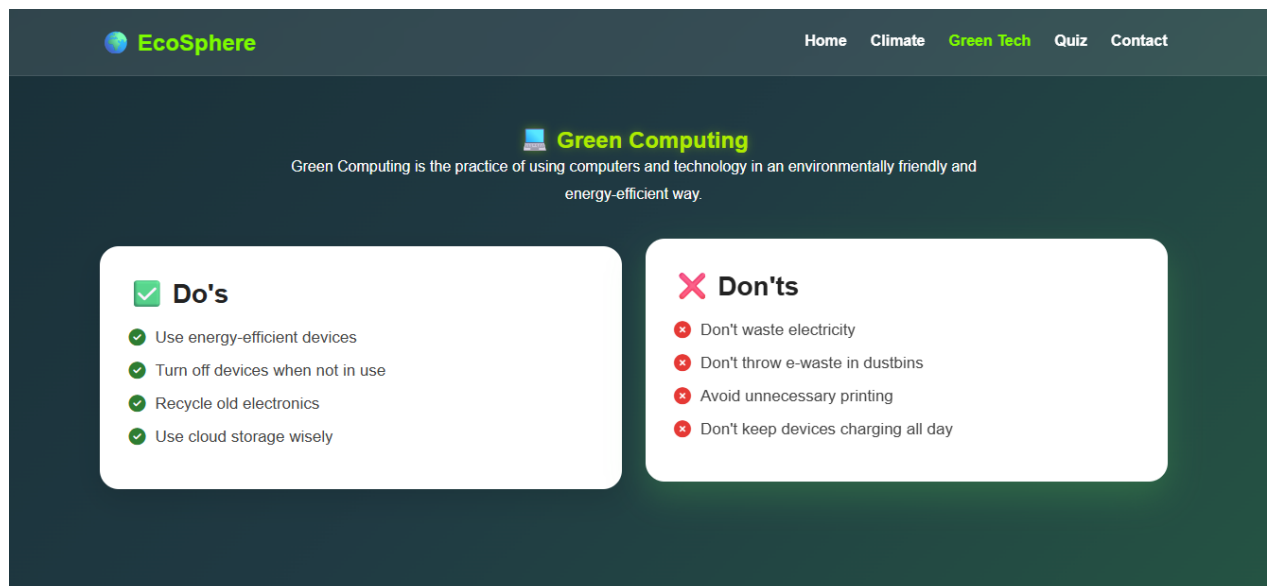


Figure 5: Carbon Footprint Calculator

[Home](#) [Climate](#) [Green Tech](#) [Quiz](#) [Contact](#)



Carbon Footprint Calculator

Estimate your daily carbon footprint and learn how eco-friendly your lifestyle is.

Calculate

Figure 6: Contact and Footer

[Home](#) [Climate](#) [Green Tech](#) [Quiz](#) [Contact](#)



Contact Us

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EcoSphere

Green Computing for a Sustainable Future

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CONCLUSION

The Ecosphere – Green Computing project was successfully developed to promote environmental awareness and sustainable computing practices. The website provides useful information about climate change, green technology, and green computing in a simple and user-friendly manner.

The project also includes a Carbon Footprint Calculator that helps users estimate their carbon emissions and encourages them to adopt eco-friendly habits. By using modern web technologies such as HTML5, CSS3, and JavaScript, the website is responsive, interactive, and easy to use.

Overall, this project highlights the importance of protecting the environment through responsible use of technology and motivates users to contribute towards a cleaner, greener, and more sustainable future.

REFERENCES

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